LAB 10: PAGING TECHNIQUE

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C PROGRAM

```
#include <stdio.h>
int main() {
    int pageTable[20];
    int pageSize;
    int numPages;
    int logicalAddress;
    int pageNumber;
    int offset;
    int frameNumber;
    int physicalAddress;
    int i;

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);
    printf("Enter Number of Pages: ");
    scanf("%d", &numPages);
    printf("Enter Frame Numbers for Each Page:
");
    for(i = 0; i < numPages; i++)
    {
        printf("Page %d -> Frame: ", i);
        scanf("%d", &pageTable[i]);
    }
    printf("Enter Logical Address: ");
    scanf("%d", &logicalAddress);

    pageNumber = logicalAddress / pageSize;
    offset = logicalAddress % pageSize;

    if(pageNumber >= numPages)
    {
        printf("Invalid Logical Address
");
        return 0;
    }

    frameNumber = pageTable[pageNumber];
    physicalAddress = (frameNumber * pageSize) + offset;

    printf("
Page Number: %d", pageNumber);
    printf("
Offset: %d", offset);
    printf("
Frame Number: %d", frameNumber);
    printf("
Physical Address: %d
", physicalAddress);
    return 0;
}
```

SHELL SCRIPTING

```
#!/bin/bash
echo "Enter Page Size:"
read pageSize
echo "Enter Number of Pages:"
read numPages
declare -a pageTable
for ((i=0; i<numPages; i++))
do
    echo "Enter Frame Number for Page $i:"
    read pageTable[$i]
done
echo "Enter Logical Address:"
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
    echo "Invalid Logical Address"
    exit
fi
frameNumber=${pageTable[$pageNumber]}
physicalAddress=$((frameNumber * pageSize + offset))
echo "Page Number: $pageNumber"
echo "Offset: $offset"
echo "Frame Number: $frameNumber"
echo "Physical Address: $physicalAddress"
```

OUTPUT

```
┌──(kali㉿Harish)-[~]
└─$ nano paging.sh
┌──(kali㉿Harish)-[~]
└─$ chmod +x paging.sh
┌──(kali㉿Harish)-[~]
└─$ bash paging.sh
Enter Page Size:
2
Enter Number of Pages:
3
Enter Frame Number for Page 0:
1
Enter Frame Number for Page 1:
2
Enter Frame Number for Page 2:
3
Enter Logical Address:
4
Page Number: 2
Offset: 0
Frame Number : 3
Physical Address : 6
```